

Aejandro De La Torre

LIS 440

wk15: More Graphing with Pandas/Plotting - State Data

12/16/25

Code

```
# Import the necessary packages
import pandas as pd
import os
from pathlib import Path
import matplotlib.pyplot as plt

# Print version of pandas
print("Alejandro De La Torre's code....")
print(pd.__version__)

# -----
# File & Path Setup
# -----

# Set root directory (two levels up from this script)
root = Path(__file__).parent.parent.resolve()

# Show current working directory
print("Current Working Directory:", Path.cwd())

# Define key directories
data = root / 'data'
data_raw = data / 'raw'
data_processed = data / 'processed'
output = root / 'output' / 'figures'
output.mkdir(parents=True, exist_ok=True) # Make sure output folder exists

# Fetch the data
states_df = pd.read_csv(data_processed / 'YearEndStates.csv')
print("\n:Year End States Data:")
print(states_df)

# Set state and year lists
MO = "Missouri"
CA = "California"
IL = "Illinois"
ND = "North Dakota"

selected_states_3 = [MO, CA, IL]
selected_states_4 = [MO, CA, IL, ND]

years_3 = ["2010", "2015", "2020"]
years_4 = ["2010", "2015", "2020", "2024"]

# Set index to 'State'
states_df.set_index("State", inplace=True)
```

```

# =====
# Graph 1: My State, All Years
# =====
states_df.loc[M0].plot(kind="bar", title=f"{M0}, All Years")
plt.xlabel("Years")
plt.ylabel("Avg Home Price (USD)")
plt.tight_layout()
plt.savefig(output / "graph1_M0_all_years.png")
plt.close()

# =====
# Graph 2: Another State, All Years
# =====
states_df.loc[CA].plot(kind="bar", title=f"{CA}, All Years")
plt.xlabel("Years")
plt.ylabel("Avg Home Price (USD)")
plt.tight_layout()
plt.savefig(output / "graph2_CA_all_years.png")
plt.close()

# =====
# Graph 3: All States, Year = 2009
# =====
# I emailed professor Nyhoff to ask about why it says 2008 when data only goes back to
2009
states_df["2009"].plot(kind="bar", title="2009, All States")
plt.xlabel("States")
plt.ylabel("Avg Home Price (USD)")
plt.tight_layout()
plt.savefig(output / "graph3_all_states_2009.png")
plt.close()

# =====
# Graph 4: All States, Year = 2020
# =====
states_df["2020"].plot(kind="bar", title="2020, All States")
plt.xlabel("States")
plt.ylabel("Avg Home Price (USD)")
plt.tight_layout()
plt.savefig(output / "graph4_all_states_2020.png")
plt.close()

# =====
# Graph 5: My State, Selected Years
# =====
states_df.loc[M0, years_3].plot(kind="bar", title=f"{M0}, Selected Years")
plt.xlabel("Years")
plt.ylabel("Avg Home Price (USD)")
plt.tight_layout()
plt.savefig(output / "graph5_M0_selected_years.png")
plt.close()

```

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# =====
# Graph 6: Other State, Selected Years
# =====
states_df.loc[CA, years_3].plot(kind="bar", title=f"{CA}, Selected Years")
plt.xlabel("Years")
plt.ylabel("Avg Home Price (USD)")
plt.tight_layout()
plt.savefig(output / "graph6_CA_selected_years.png")
plt.close()

# =====
# Graph 7: Both States, All Years Clustered
# =====
states_df.loc[[MO, CA]].transpose().plot(kind="bar", title="Selected States, All Years Clustered")
plt.xlabel("Years")
plt.ylabel("Avg Home Price (USD)")
plt.tight_layout()
plt.savefig(output / "graph7_both_states_all_years_clustered.png")
plt.close()

# =====
# Graph 8: All Years, Both States Clustered
# =====
ax = states_df.loc[[MO, CA]].plot(kind="bar", title="All Years, Selected States Clustered")
plt.xlabel("States")
plt.ylabel("Avg Home Price (USD)")
plt.legend(title="Year", bbox_to_anchor=(1.05, 1), loc='upper left') # Move legend
outside plot
plt.tight_layout()
plt.savefig(output / "graph8_all_years_both_states_clustered.png", bbox_inches='tight')
# Adjust bounding box
plt.close()

# =====
# Graph 9: 3 States x 4 Years Clustered
# =====
states_df.loc[selected_states_3, years_4].transpose().plot(kind="bar", title="3 States x 4 Years Clustered")
plt.xlabel("Years")
plt.ylabel("Avg Home Price (USD)")
plt.tight_layout()
plt.savefig(output / "graph9_3_states_4_years.png")
plt.close()

# =====
# Graph 10: 4 States x 3 Years Clustered
# =====
states_df.loc[selected_states_4, years_3].transpose().plot(kind="bar", title="4 States x 3 Years Clustered")
plt.xlabel("Years")
plt.ylabel("Avg Home Price (USD)")

```

```
plt.tight_layout()
plt.savefig(output / "graph10_4_states_3_years.png")
plt.close()
```

Shell

```
>>> %Run more_viz.py
Alejandro De La Torre's code....
2.3.3
Current Working Directory: /Users/alexdelatorre/Library/CloudStorage/OneDrive-UW-Madison/Coursework/L I S
440/Lectures/Week 15/code

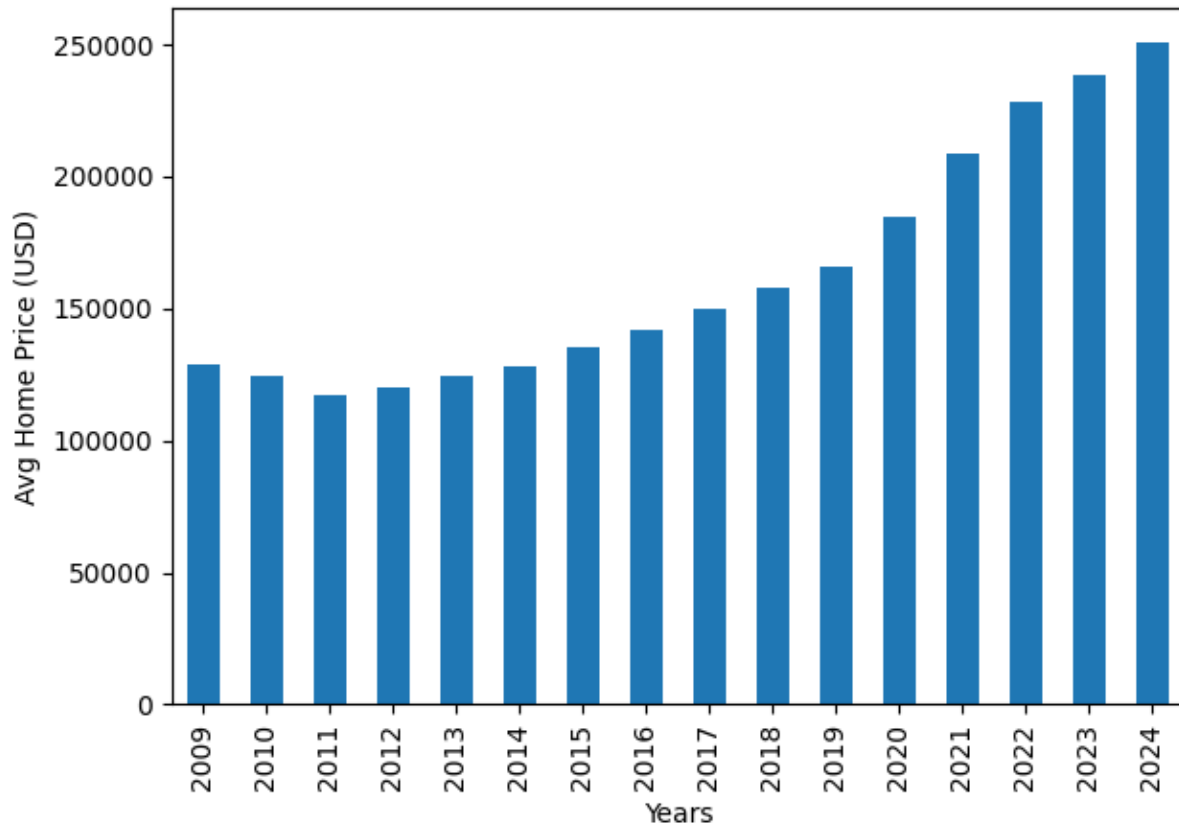
:Year End States Data:
      State   2009   2010 ...   2022   2023   2024
0  California 311023.53 308040.15 ... 758754.12 784787.20 814869.24
1    Texas 137792.11 135415.93 ... 303016.77 299382.34 305584.58
2   Florida 157574.39 146743.99 ... 399559.08 407854.14 411048.34
3   New York 230031.14 228144.51 ... 399275.50 424363.80 457969.41
4 Pennsylvania 158216.69 156238.59 ... 245453.87 255462.76 271131.94
5   Illinois 164347.91 154335.41 ... 240723.10 253570.28 274672.45
6    Ohio 117075.28 113819.48 ... 206611.81 217704.71 233484.84
7    Georgia 139092.96 129692.98 ... 312640.11 322212.12 332569.71
8 North Carolina 159771.99 154752.41 ... 313530.44 323426.08 333886.44
9   Michigan 99794.06 95462.49 ... 219840.31 229346.62 244017.34
10 New Jersey 310271.35 300019.93 ... 474145.56 513535.78 562751.82
11  Virginia 220621.26 217374.14 ... 362770.93 381549.17 404080.51
12 Washington 269150.31 252682.96 ... 582236.52 588218.91 617896.54
13  Arizona 166275.89 151306.54 ... 438390.23 435431.36 442034.90
14 Massachusetts 308628.67 308399.39 ... 569241.96 613340.08 655451.00
15  Tennessee 136303.92 134085.76 ... 298251.17 309688.26 318089.17
16  Indiana 113610.32 111914.88 ... 220701.77 230261.32 242601.89
17  Maryland 267723.95 256284.75 ... 402707.06 417044.25 435842.92
18  Missouri 128499.25 124400.05 ... 228586.44 238290.44 250999.17
19 Wisconsin 159826.87 154089.33 ... 270329.19 288366.99 311931.58
20  Colorado 227592.19 222762.04 ... 552886.77 551337.36 568130.82
21  Minnesota 174241.90 167875.50 ... 322986.60 328722.56 341696.19
22 South Carolina 147466.98 143390.23 ... 280273.06 288320.49 299251.37
23  Alabama 124185.63 119078.48 ... 214513.61 218316.75 224422.21
24  Louisiana 150336.71 150064.90 ... 202011.66 196182.49 200349.79
25  Kentucky 104922.45 104117.09 ... 188238.72 196081.51 211514.78
26  Oregon 230121.27 216014.06 ... 493347.96 494774.83 509359.16
27  Oklahoma 109426.58 108587.82 ... 193582.46 199578.87 207682.05
28 Connecticut 270042.35 261377.63 ... 367995.25 405716.53 436979.81
29  Utah 211402.14 197816.05 ... 520862.97 519653.42 538612.74
30  Iowa 119097.98 118033.32 ... 205448.96 211996.33 221271.26
31  Nevada 163757.89 152483.41 ... 443237.45 438564.51 464226.92
32  Arkansas 115719.57 113960.53 ... 193485.87 198227.73 209125.77
33  Mississippi 111036.07 108038.88 ... 172047.79 170806.36 177947.39
34  Kansas 118491.46 116284.71 ... 208694.04 217680.00 230577.90
35  New Mexico 175281.03 165914.70 ... 279280.43 292675.45 306808.18
36  Nebraska 127499.99 124925.70 ... 239482.97 250554.16 263452.86
37  Idaho 166362.62 150087.25 ... 452650.43 443585.62 460946.52
38  West Virginia 97626.64 97685.54 ... 150197.89 154625.99 165201.42
39  Hawaii 466362.83 485490.71 ... 973411.46 966682.60 990827.06
```

40	New Hampshire	223059.44	219490.07	...	438212.09	473180.38	505056.34
41	Maine	176486.88	173285.26	...	349082.24	378044.49	395665.80
42	Rhode Island	229453.63	227130.02	...	415484.17	446108.45	483502.02
43	Montana	190006.06	185568.21	...	438647.49	448230.07	457754.10
44	Delaware	238111.20	229386.31	...	354147.36	371266.57	388017.14
45	South Dakota	142408.38	143231.23	...	289091.68	291921.41	306833.33
46	North Dakota	159386.74	161746.16	...	256450.07	252989.26	270824.88
47	Alaska	245780.69	246957.24	...	359764.67	361470.99	383213.96
48	District of Columbia	381853.59	388208.06	...	733018.84	721522.80	730824.30
49	Vermont	204896.72	205821.08	...	355848.04	377678.59	396564.35
50	Wyoming	195996.86	195026.42	...	324116.17	333997.57	351878.71

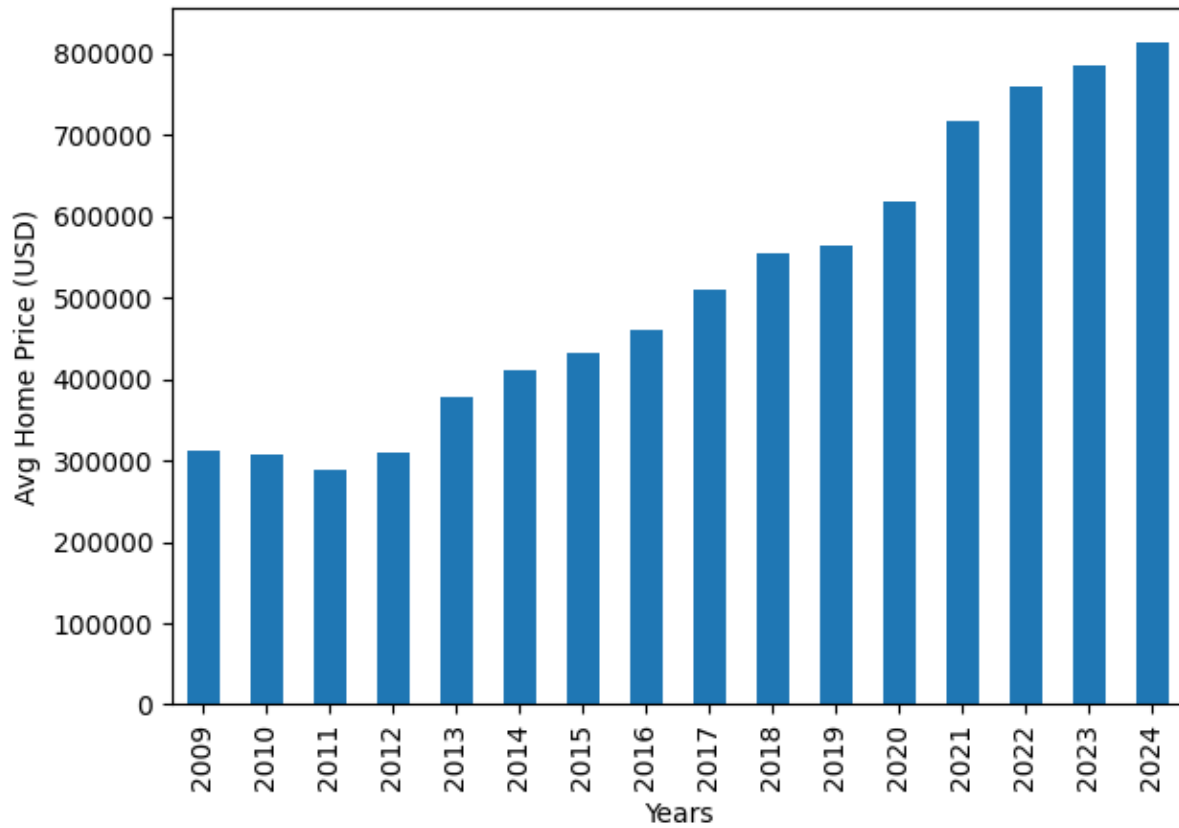
[51 rows x 17 columns]

>>>

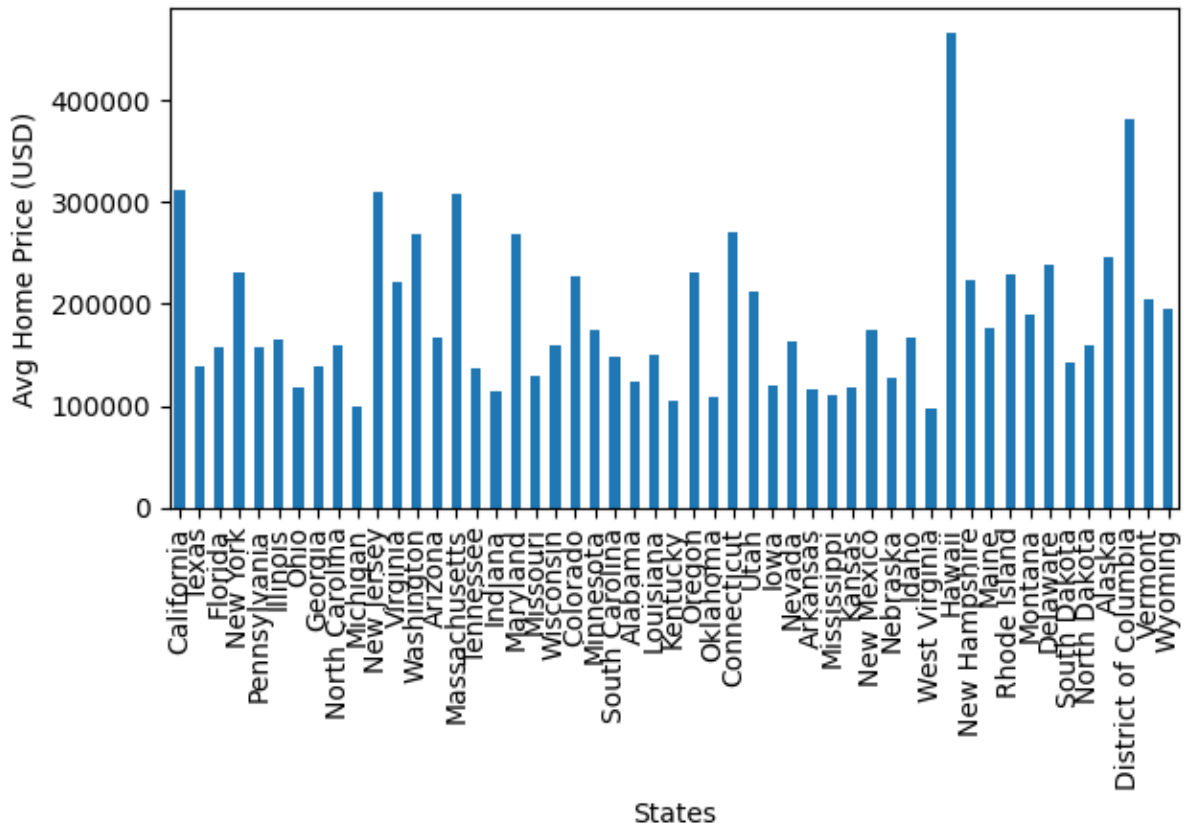
Missouri, All Years



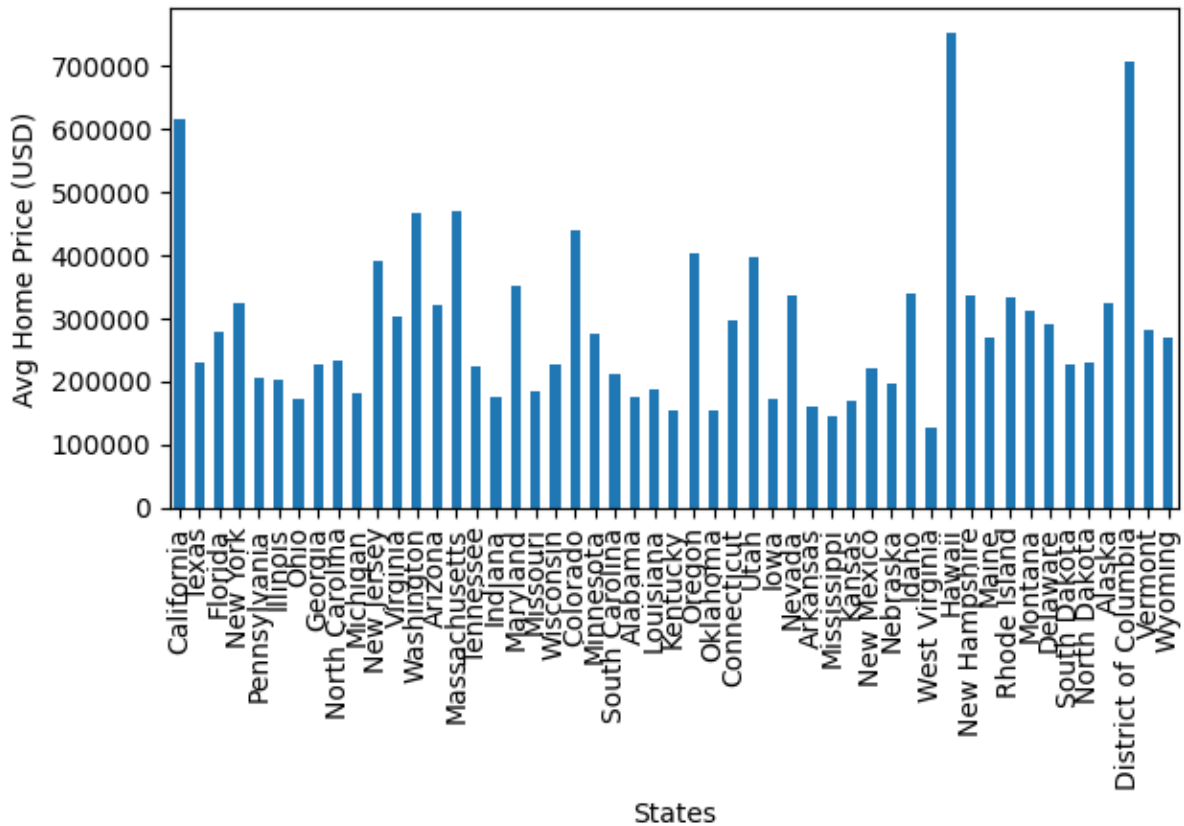
California, All Years



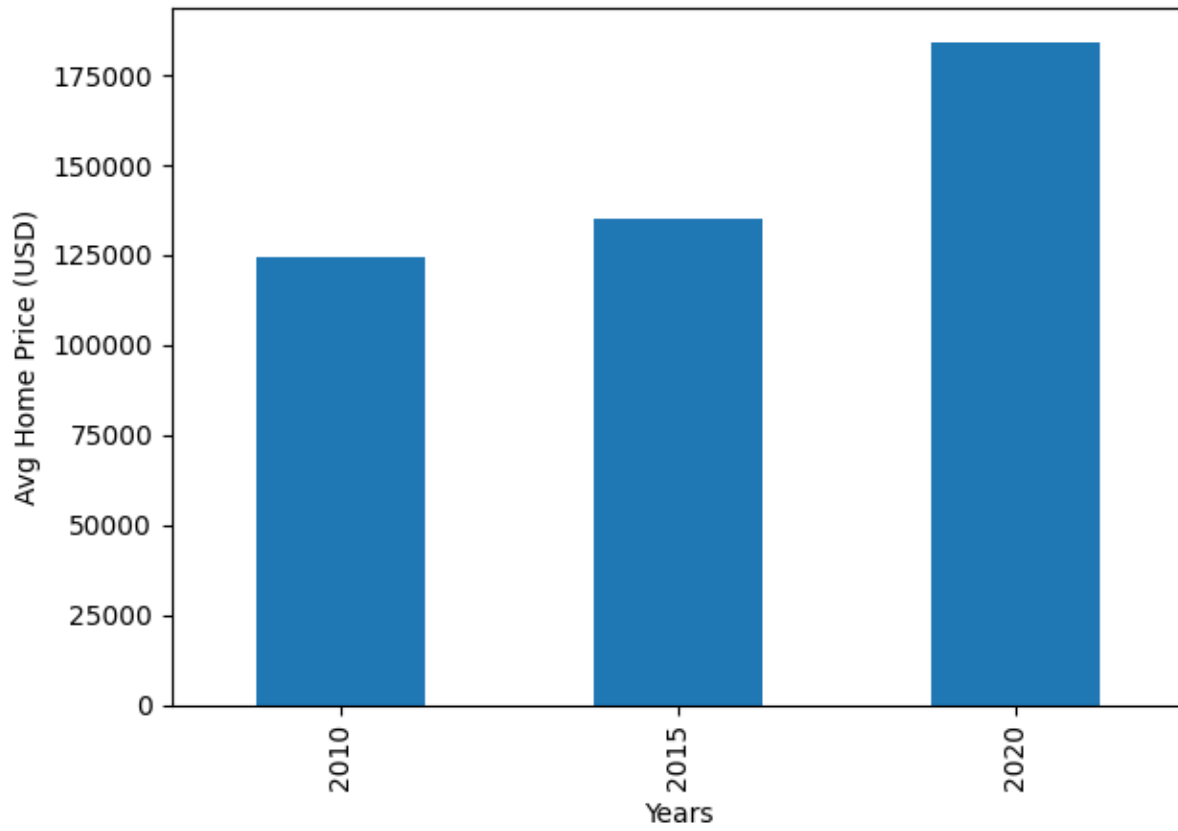
2009, All States



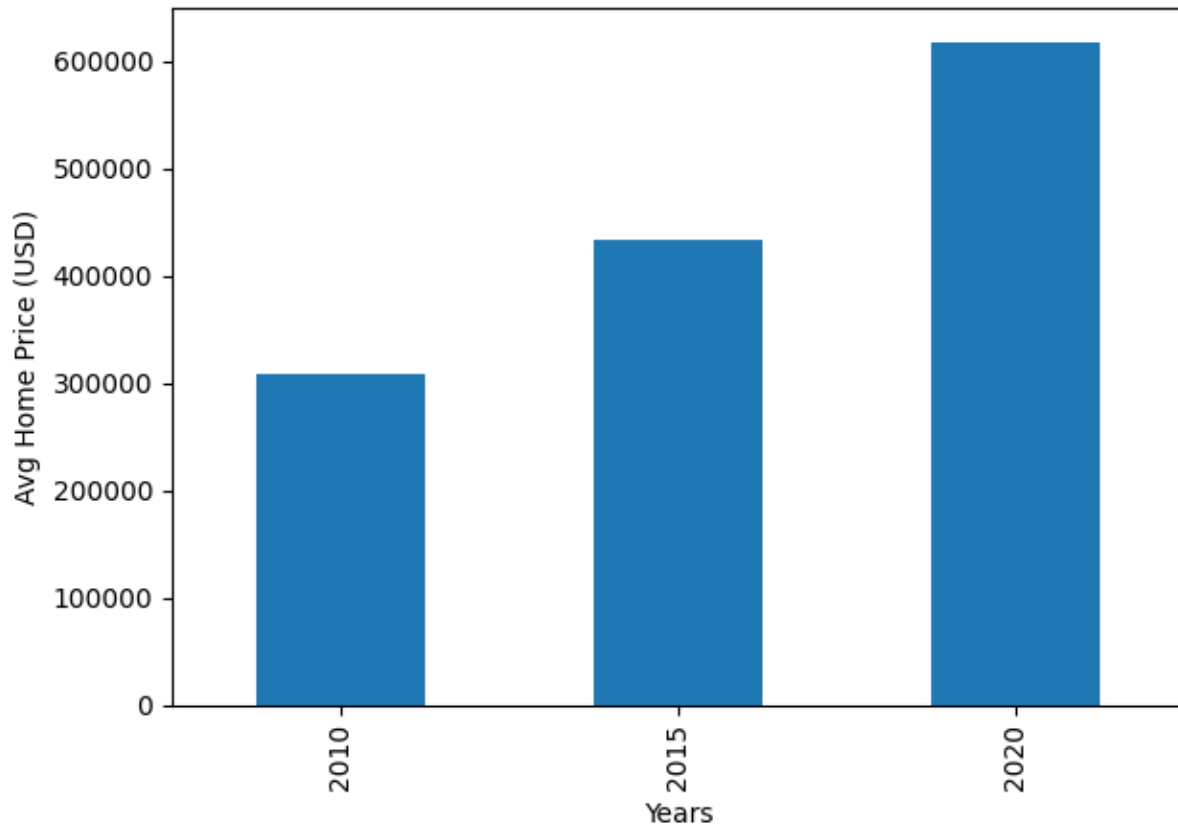
2020, All States



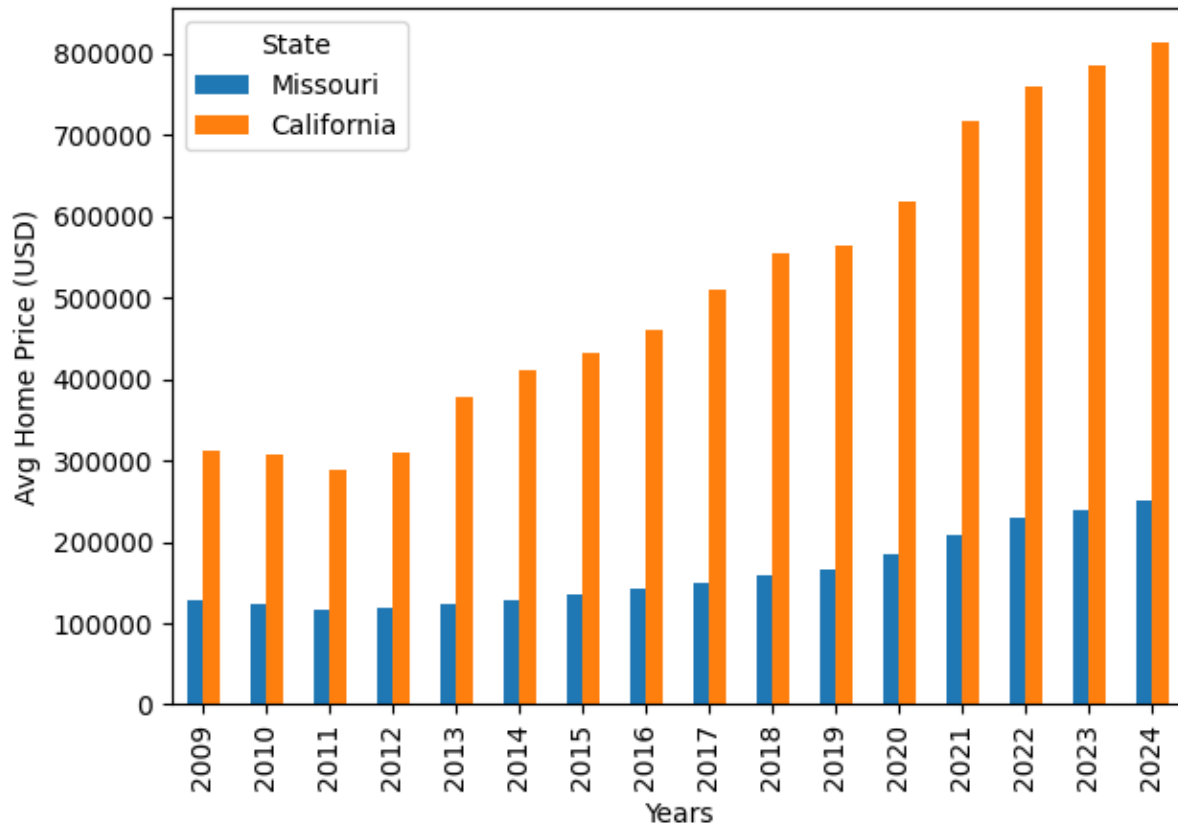
Missouri, Selected Years



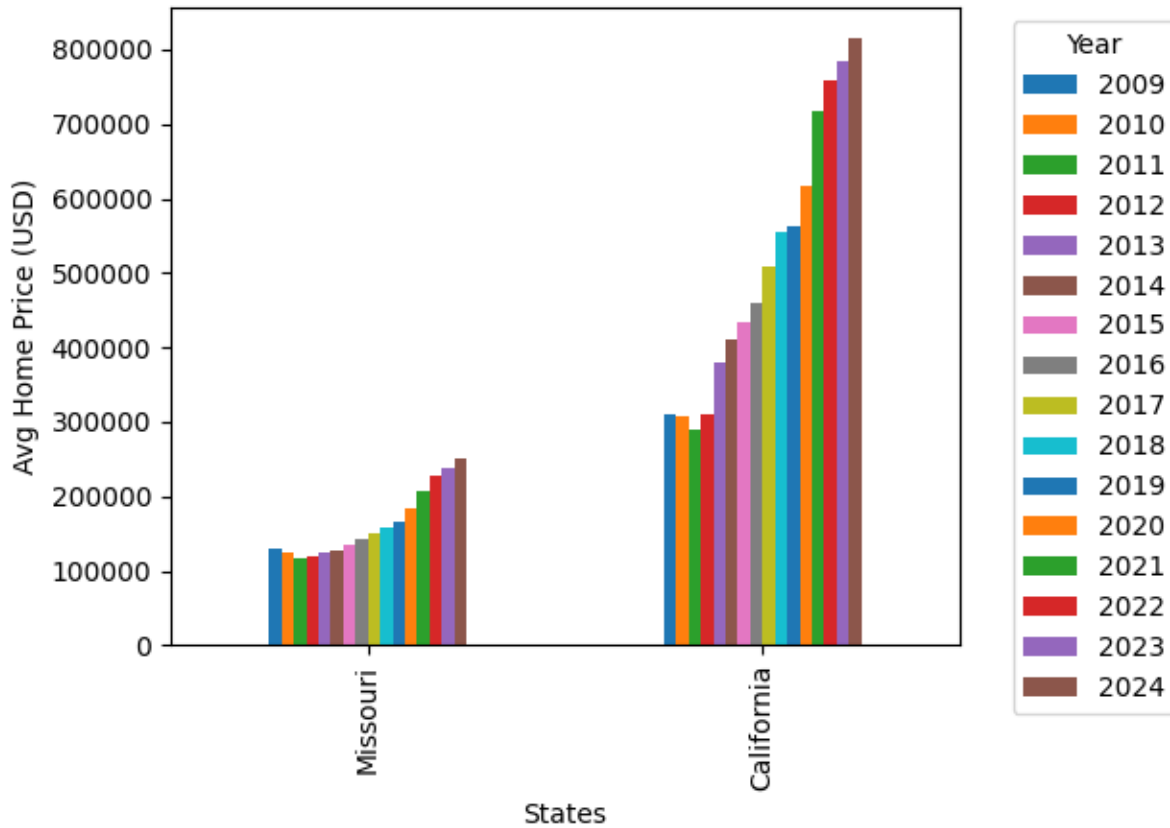
California, Selected Years



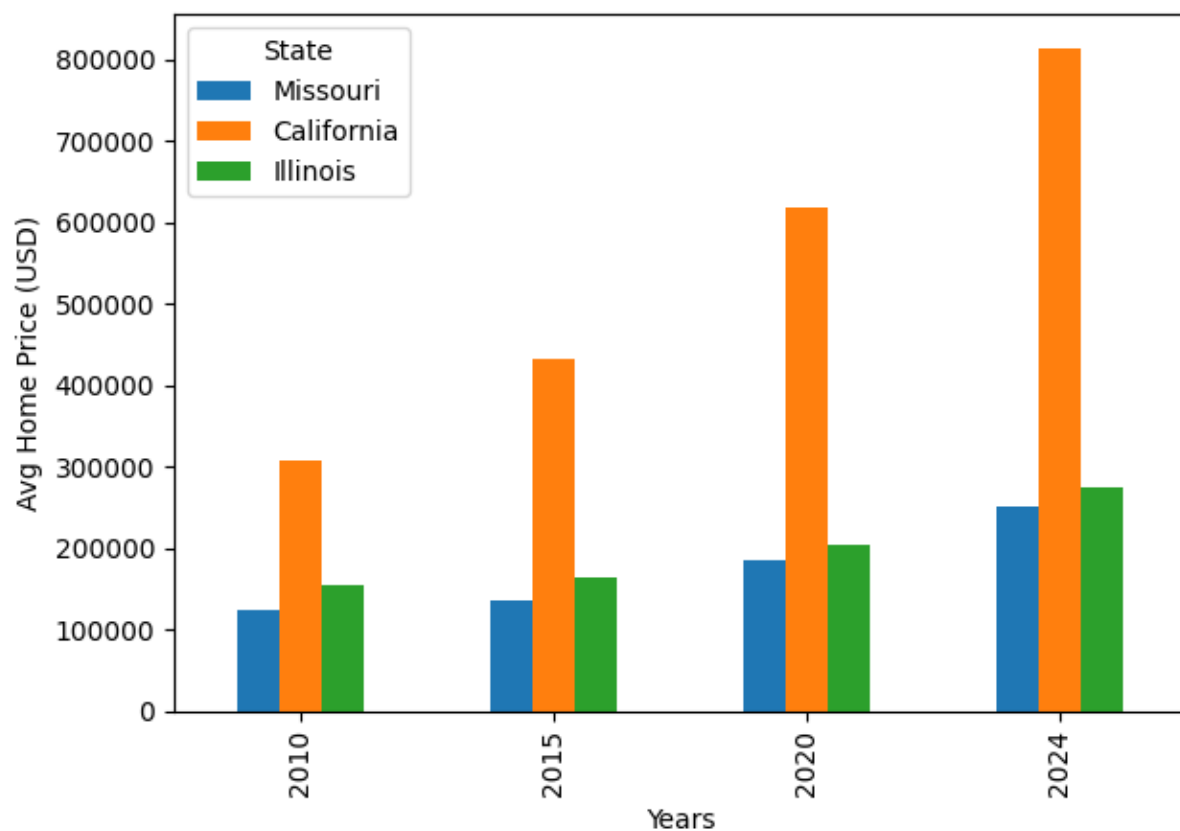
Selected States, All Years Clustered



All Years, Selected States Clustered



3 States x 4 Years Clustered



4 States x 3 Years Clustered

